Mississippi Valley State University

CS 341 — Discrete Structures

Practice Set 19

One-to-One and Onto Functions

Covers Chartrand & Zhang, Section 5.4

Status of this set

Ungraded practice. Solutions released with it. Write proofs in complete sentences.

Part I: How to Practice This Material

Four jobs, four templates

Know which one you are doing

	To prove	To disprove
One-to-one	Assume $f(a) = f(b)$; derive $a = b$.	Name $a \neq b$ with $f(a) = f(b)$.
Onto	Take arbitrary b; produce a with $f(a) = b$.	Name a b with no preimage.

Scratch work runs backwards

To prove onto you must *produce* a preimage. Find it by solving $f(x) = b$ for x on scratch paper, then write the proof forwards: "Let b be given. Set $a = \ldots$ Then $f(a) = b$."
The finished proof hides the search. That is normal, and expected of you too.

Three mistakes to expect

1. **Starting a one-to-one proof from $a \neq b$.** Use the contrapositive form.
2. **Asserting onto without producing a preimage.** "Every value is hit" is a claim, not a proof.
3. **Ignoring the codomain.** The same rule can be onto or not depending on it.

Part II: Practice Problems

Section A — Explicit functions

1. Let $A = \{1, 2, 3\}$ and $B = \{p, q, r, s\}$. For each function, state whether it is one-to-one, onto, both, or neither, with witnesses for failures.

(a) $f = \{(1, p), (2, q), (3, r)\}$

(b) $g = \{(1, p), (2, p), (3, s)\}$

(c) $h : B \to A$ with $h = \{(p, 1), (q, 2), (r, 3), (s, 1)\}$

Section B — Proving

2. Prove that $f : \mathbb{R} \to \mathbb{R}$ defined by $f(x) = 7x + 2$ is bijective, and find f^{-1}.

3. For each function, determine whether it is one-to-one and whether it is onto. Justify.

(a) $f : \mathbb{Z} \to \mathbb{Z}$, $f(n) = n + 5$.

(b) $g : \mathbb{Z} \to \mathbb{Z}$, $g(n) = 3n$.

(c) $h : \mathbb{R} \to \mathbb{R}$, $h(x) = x^3 - x$.

Section C — The codomain matters

4. Consider the rule $f(x) = x^2$ in three settings. For each, state whether it is one-to-one and whether it is onto.

(a) $f : \mathbb{R} \to \mathbb{R}$

(b) $f : \mathbb{R} \to [0, \infty)$

(c) $f : [0, \infty) \to [0, \infty)$

Section D — Writing

5. In five to eight complete sentences, explain why the standard way to prove a function is one-to-one starts from $f(a) = f(b)$ rather than from $a \neq b$. Name the result from earlier in the term that justifies the switch, and say what practical advantage it gives.

Section E — Challenge (optional)

6. Let A be a finite set and $f : A \to A$ a function. Prove that f is one-to-one if and only if f is onto. Then give an example showing this fails for infinite sets.